

```
In [1]: # WAP take number1= 100 and number2=200 perform addition operation
number1=100 # int
number2=200
add=number1+number2
print(f"the addition of {number1} and {number2} is {add}")
```

the addition of 100 and 200 is 300

```
In [ ]: # this 100 we enter manually
# we can take these values from keyboard
```

input

```
In [7]: input("enter the value:")
```

enter the value:100

Out[7]: '100'

```
In [ ]: when we take the values using input keyword
the type of the value is string
```

```
In [15]: number1=input("enter number1:") # ask the n1='100'
number2=input("enter number2:") # ask the n2='200'
print(type(number1))
print(type(number2))
add=number1+number2 # '100'+ '200'
add # 100200
```

enter number1:apple
enter number2:200
<class 'str'>
<class 'str'>

Out[15]: 'apple200'

```
In [13]: '100'+ '200'
```

Out[13]: '100200'

```
In [1]: input()

# whenever we provides anything value /string that is str data type
```

10

Out[1]: '10'

```
In [3]: # i want to add two numbers , from keyboard
number1=input("enter number1:")
number2=input("enter number2:")
number1+number2

# step-1: input()==== number1='100'
# step-2: input()==== number2='200'
# step-3: '100'+'200'='100200'
```

```
enter number1:100
enter number2:200
```

Out[3]: '100200'

```
In [5]: int(number1)
```

Out[5]: 100

```
In [6]: # Method-1
number1=input("enter number1:") # number1='100'
number2=input("enter number2:") # number2='200'
int(number1)+int(number2)       # int('100')+int('200')== > 100+200=300
```

```
enter number1:100
enter number2:200
```

Out[6]: 300

```
In [8]: number1=int('100')
number2=int('200')
number1+number2
```

Out[8]: 300

```
In [9]: number1='100'
number2='200'
add=number1+number2
int(add)
```

Out[9]: 100200

```
In [10]: # Method-2
number1=int(input("enter number1:")) # number1=int('100')
number2=int(input("enter number2:")) # number2=int('200')
add=number1+number2
add
```

```
enter number1:200
enter number2:300
```

Out[10]: 500

```
In [14]: # Method-1
print("====Method1====")
number1=input("enter number1:") # number1='100'
number2=input("enter number2:") # number2='200'
add=int(number1)+int(number2)
print(f"the addition of {number1} and {number2} is {add}")

# Method-2
print("====Method2====")
number1=int(input("enter number1:")) # number1=int('100')
number2=int(input("enter number2:")) # number2=int('200')
add=number1+number2
print(f"the addition of {number1} and {number2} is {add}")

# Method-3
print("====Method-3====")
number1=input("enter number1:") # number1='100'
number2=input("enter number2:") # number2='200'
print(f"the addition of {number1} and {number2} is {int(number1)+int(number2)}")

====Method1====
enter number1:100
enter number2:200
the addition of 100 and 200 is 300
====Method2====
enter number1:300
enter number2:400
the addition of 300 and 400 is 700
====Method-3====
enter number1:500
enter number2:600
the addition of 500 and 600 is 1100
```

```
In [ ]: #Q)WAP ask the user
# enter your name
# name=input("enter your name")
# enter your age
# age = int(input("enter your age"))
# enter your city
# city=input("enter your city")
# enter your salary
# salary= float(input('enter the salary'))

# print("My name is python, Im 10years old, I came from world, I earned 100")
```

```
In [16]: name="python"
name
```

```
Out[16]: 'python'
```

```
In [18]: name=input("enter your name:")
age=int(input("enter your age:"))
city=input("enter your city:")
salary=float(input("enter your salary:"))
print(f"My name is {name}, I'm {age} years old, I came from {city}, I earned {salary}")
```

enter your name:python

enter your age:10

enter your city:world

enter your salary:100000

My name is python, I'm 10 years old, I came from world, I earned 100000.0

```
In [20]: # I want to take float values and integer values to add
n1=float(input("enter a number1:"))
n2=float(input("enter a number2:"))
```

enter a number1:100

enter a number2:100.5

eval: evaluate

- eval is used for instead of int and float
- we no need to worry which data type conversion need to apply
- if you provide integer values, then eval will convert into integer only
- if you provide float values, then eval will convert float type only
- eval will fail if you provide string

```
In [24]: print("====Method1====")
number1=eval(input("enter number1:")) # number1='100'
number2=eval(input("enter number2:")) # number2='200'
add=number1+number2
print(f"the addition of {number1} and {number2} is {add}")
```

====Method1====

enter number1:100

enter number2:200.5

the addition of 100 and 200.5 is 300.5

```
In [ ]: # shift+enter is used to run the cell
```

```
In [ ]: #Q) wap ask the user enter two numbers
# add
# sub
# mul
# div
```

```
In [26]: num1=eval(input("Enter value1: "))
num2=eval(input("Enter value2: "))
print(f"additon is: {num1+num2}")
print(f"Subtraction is: {num1-num2}")
print(f"Multiplication is: {num1*num2}")
print(f"Divison is: {num1/num2}")
```

```
Enter value1: 15
Enter value2: 7
additon is: 22
Subtraction is: 8
Multiplication is: 105
Divison is: 2.142857142857143
```

round

```
In [31]: round(2.142857142857143,3)
```

```
Out[31]: 2.143
```

```
In [32]: # WAP ask the user enter three numbers
# find the average

num1 = eval(input("enter first number:"))
num2 = eval(input("enter second number:"))
num3 = eval(input("enter third number:"))
summ=num1+num2+num3
avg=round(summ/3,2)
print(f"Average of {num1} and {num2} and {num3} is {avg}")
```

```
enter first number:10
enter second number:20
enter third number:30
Average of 10 and 20 and 30 is 20.0
```

```
In [ ]: # WAP ask the user enter income
# ask the user how much tax : eval(input("enter tax in percentage"))=10
# calculate how much tax you want to pay

100000*10/100
```

```
In [34]: income=eval(input("Enter Income : "))
per=eval(input("Enter tax in percentage : "))
calc=income*per/100
print(f"percentage is{calc}")
```

```
Enter Income : 100000
Enter tax in percentage : 10
percentage is10000.0
```

```
In [35]: # WAP ask the user enter income
income=eval(input("enter income:"))
#ask the user how much tax : eval(input("enter tax in percentage"))=10
tax_per=eval(input("enter tax in %:"))
#calculate how much tax you want to pay
tax=income*tax_per/100
tax
```

```
enter income:10000
enter tax in %:10
```

Out[35]: 1000.0

```
In [ ]: # 2000
# tip 10 percentage on bill
# total amount=2200

# assume that you went to a restuarent
# you made a bill of 2000 rs
# you also want to pay 10% tip on the bill
# now calculate total bill
# tip=2000*10/100
# total bill= 2000+ (tip)
```

```
In [ ]: project1
project1_test
project1_test1
project1_test_final
```

```
In [36]: bill = eval(input("enter your bill:"))
tip_per = eval(input("enter tip percentage:"))
tip_amout = bill*tip_per/100
total_bill = bill+tip_amout
print(f"User has to pay tip is {tip_amout}")
print(f"User has to pay total bill is {total_bill}")
```

```
enter your bill:2000
enter tip percentage:10
User has to pay tip is 200.0
User has to pay total bill is 2200.0
```

time

```
In [37]: num1=eval(input("enter number1:"))
num2=eval(input("enter number2:"))
```

```
enter number1:10
enter number2:20
```

```
In [41]: import time

name1='python'
name2='hello'
print(name1)
time.sleep(5)
print(name2)
```

```
python
hello
```

```
In [39]: import time
```

```
In [40]: dir(time)
```

```
Out[40]: ['_STRUCT_TM_ITEMS',
          '__doc__',
          '__loader__',
          '__name__',
          '__package__',
          '__spec__',
          'altzone',
          'asctime',
          'ctime',
          'daylight',
          'get_clock_info',
          'gmtime',
          'localtime',
          'mktime',
          'monotonic',
          'monotonic_ns',
          'perf_counter',
          'perf_counter_ns',
          'process_time',
          'process_time_ns',
          'sleep',
          'strftime',
          'strptime',
          'struct_time',
          'thread_time',
          'thread_time_ns',
          'time',
          'time_ns',
          'timezone',
          'tzname']
```

```
In [ ]: CTRL+P ===== print pages
```

```
In [ ]:
```

```
In [ ]:
```

```
In [ ]:
```

